Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Martijn Hoogeland	Email	Martijn.hoogeland@tno.nl
Institute Name	TNO			
Address	Molengraaffsingel 8, 2629 JD Delft			

2. Technology Overview

Title of Proposed Technology	Advanced instruments and detection methodology for hydrogen leakage.			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (X), Hydrogen (X), Cybersecurity ( ), AI (X), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	electrical * Electronics		Mechanical * materials	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Optical measuring Instruments	Shipbuilding/ Marine systems	Other Shipbuilding/Marine System Technologies
Concept of Proposed Technology	This proposal brings in three aspects that could also be pursued separately. 1) Is about low-cost sensors for hydrogen detection, 2) is about the data collection and interpretation module which gives advice and 3) is about the user-friendly interaction to lower any barrier to work fluently with the system and hydrogen applications. Small, easy to manufacture and install optics-based sensors combined with reliable, real-time interpretation of the data will enable operators to take the most appropriate response to hydrogen leakage. This proposal combines innovative low-cost sensors and AI driven, physics based, data-to-advice algorithms to accurately identify a leakage and the source. This accelerates the implementation of hydrogen in energy applications such as marine and road powertrains, electrolysers, stationary fuel cells and pipelines by providing reliable, effective and low-cost detection systems. Two specific applications stand out: detecting internal leakage in electrolysers and leakage detection for storage systems for mobile applications.			

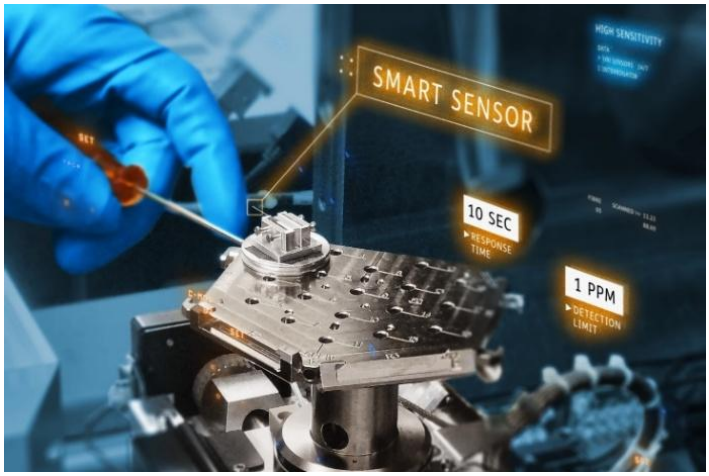
	Real-time hydrogen gas concentration data are obtained through optics-based sensors [3, 4]. The collected data are leveraged to train AI models, allowing extensive multi-point datasets to be accumulated under various parameters (leak and initial conditions, physical characteristics) By utilizing a database accumulated through AI models, this approach enables the automatic identification of hydrogen leakage points when hydrogen gas is detected under various conditions during hydrogen leak accidents. We aim to develop a 3D GUI-based monitoring system that applies the proposed technology to hydrogen storage and supply facilities as well as hydrogen-powered vessels, enabling operators and managers to respond quickly to incidents. In addition, the system is intended to accurately control hydrogen leakage and establish a foundation for minimizing accident-related damage.  *Figure 1. TNO optical fiber hydrogen sensor lab demonstration.*
Rationale for Joint R&D Project	Brings together a unique combination of knowledge & capabilities on sensors, AI and application specific requirements. This means the collaboration amongst disciplines, and with the best members from both South Korea and The Netherlands. - Green H_2 product system design & engineering At TNO, we have capabilities for full system engineering and R&D facility including critical data and analysis/diagnostic modelling capability for H_2 industry. - Measurement and monitoring instruments TNO has capabilities for the development of superior quality sensors (OFS and PIC) design and R&D for high tech sector (energy, space, semicon, etc.) over 40 years. South Korea is expected to have industrial potential to manufacture such sensors at high quality, large quantity and low cost. - Application in road, heavy duty and maritime TNO is involved in application cases for road, heavy duty, maritime and offshore. As such, we analyse, test and innovate for the hydrogen application for maritime and heavy-duty industry. Probabilistic methods, combining physics-based models with AI are used. Physical experimental facilities are available to test proposed advanced leak detection. South Korean companies are expected to bring in specific cases, along superior engineering power from for instance the shipbuilding industry and heavy duty truck industry.

	 Figure 3. TNO test lab for sustainable maritime engines.
Key Words	1. Photonics and optical technologies 2. Digital and information technologies 3. Industrial safety 4. Maritime safety 5. AI detection technologies 6. 3D GUI monitoring system.

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	1. Define suitable methodology for reliable and cost-effective hydrogen leakage sensing 2. Develop low-cost, reliable instrumentation 3. Design a model supported data-to-advice (DTA) system based on multi-point measurement 4. Automatic hydrogen leak point detection algorithms based on a database (multi-point hydrogen gas concentration) accumulated using AI models 5. Development of a 3D GUI Monitoring System for Hydrogen Control and Damage Minimization *Figure 2. TNO Faraday lab with test bench for PEM electrolyser cell/stack.*			
R&D Target Specifications	1. The methodology shall include secondary parameters such as temperature, pressure, to avoid the need of many hydrogen sensors 2. Individual instruments (hydrogen sensors) will have a unit cost of less than 1 $ and Mean Time Between Failures (MTBF) of 5 years 3. A typical measurement and detection system shall cost 40% less than the current equivalent 4. Promotion of research outcomes through the application of optics-based sensor–driven AI safety technologies to establish and generalize a digital safety management framework			
R&D Plan	The plan below contains the proposed workplan. It is open for further detailing towards the exact use case and parties involved.			

Research Question	Actions	Deliverables and 1st planning
External leak diagnostics		
What are the system requirements for the targeted application?	• Define use case • Define approach for physical modelling of the use case. • Identification of the parameters to be measured and formal requirements	Use case description with system requirements specification M6
	• Define compliance and safety requirements • Develop integrated physical and risk model	Conceptual DFMEA/HAZOP M24

Research Question	Actions	Deliverables and 1st planning
What is the desired sensing solution?	• Conceptual system design of the multi-point and (if needed) multi-parameter sensor system	Proof-of-concept demonstration of H2 sensing M12
	• Develop, if required, (multiparameter) sensor matching the requirements • Design of distributed optic sensor system	Hydrogen (+) fibre optic sensor with required accuracy and response time M30
How to design a model supported data-to-advice (DTA) system based on multi-point measurement of hydrogen concentration?	• Assessment of potential approaches to design a DTA system integrating physical models and a Bayesian approach multiple sensing data.	Selection of a design methodology M6
	• Risk assessment based on scenarios using a physical (leakage) model • Development of a risk model which uses the measured data to improve the model (Bayesian approach) • Development of DTA system based on the requirements and compliance and safety requirements	DTA system demonstration M36
Does the system operate as required?	• Validation of the DTA system based on distributed concentration sensing in real environment.	Validated DTA system for external leakage M42
Internal leakage		
How can measurement data be mapped on off-spec physics based probabilistic models?	• Define off-spec regime and data requirements • Build off-spec physics model • Design data-to-model mapping pipeline • Validate mapping	Model outline M6
		Full model experimentally validated model M48
Which combination of parameters can provide early detection of leakage of hydrogen (through the separating membrane) for the selected use case	• Model based approach • Develop model (based on existing models) to analyse impact of pinholes (single channel), physics based • Identify through simulations how a pinhole affects parameters such as cell voltage, temperature, pressure, flow, etc. • Assess which combination of parameters can predict presence of a pinhole	Physics based model for the first estimated safety sensing requirement and safety analysis M12
	• Experimental validation • Design of experiments based on model • Development of H2 sensor • Integration of fiber optic sensor for selected multiparameter (temperature +) in lab-scale cell	Validation of the detectability at lab scale M18
How do we translate sensor and model outputs into reliable, real-time, risk-informed advice for	• Control system architecture information dimension mapping • Identify sensor and data fusion strategy	DTA development plan M12

Research Question	Actions	Deliverables and 1st planning
operators?	• Multimodal sensor system integration and sensor fusion • Advice logic, including uncertainties • Operator interface	Functional DTA demonstration M24
Does the system operate as required?	• Experimental validation at full scale	Validation of the detectability and DTA system at full scale M48
Detection diagnostics		
How can the leakage location be identified?	• Apply the hydrogen physical characteristics • Process & initial conditions according to the hydrogen leak accidents • Accumulated hydrogen gas concentration database using the AI model	Development of auto leakage location detection algorithms
How can the optimal locations for hydrogen gas leak detection be determined?	• Hydrogen dispersion range according to various leakage accidents • Identification of equipment locations based on a plot plan	Establishment of detection coverage range based on hydrogen gas concentration measurements
Does the system operate as required?	• Validation of the detection based on optics-based sensors in real environment.	Validation of the detection database using the experiment and AI model
3D GUI monitoring system		
How will the monitoring system be designed and implemented?	• Identification of equipment locations based on a plot plan • Confirmation of internal and external process procedures in real environment • Process control and real-time acquisition of hydrogen gas concentration data	

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO will provide concept development and proof of concept demonstrations. This means large part of the budget is equally shared for sensor concept development and the model development. TNO will provide its share in the AI development as well. We will further provide project management and part of the dissemination activities.
Expected R&R of Korean SME Partner	Several Korean SMEs are expected to be partners and play an active role for the following: - Provide demonstrator cases and industrial applications, such as integrity of electrolysers, safety of enclosed spaces such as fuel cell units, fuel cell rooms, fuel preparation spaces, storage/ parking lots, and fuel stations - Bring the concept of low-cost optical hydrogen sensor towards mass production - Provide monitoring and data interpretation systems towards end users, including the GUI development where co-developing the AI part is expected as well. This will include the models as developed in the project.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The implementation of low-cost, effective sensors with an AI-trained data to advise system and a user-friendly interface is expected to improve facility safety and technological capabilities for hydrogen-utilizing demand sectors in both onshore and offshore. Finally, it will support the advancement and market expansion of eco-friendly energy industries. Through the application of knowledge-based technologies in sensing and safety, this government-funded R&D is expected to address social safety issues while securing both software-based solutions and hardware technologies. Furthermore, by leveraging the acquired technologies, the scope of outcomes is anticipated to expand to products and service domains in software and hardware, thereby contributing to the development of an eco-friendly energy industry ecosystem. Examples: - Integrity of electrolysers or fuel cells can be monitored from the inside, detecting early degradation and avoid failures - Confined spaces such as fuel stations, fuel preparation rooms, parking lots are monitored and hydrogen leakages are detected prior to being an explosion risk, enhancing safety of people and operations. - The AI trained Graphical User Interface (GUI) will enhance the understanding by the users to act quickly and decisively and therefore increase the safety of hydrogen operation Technological outputs including sensor system and digital and analytical tools include: • **Validated advanced hydrogen leak detection technologies**, including high-sensitivity sensing elements and integrated detection systems capable of early leak identification under real operational conditions. • **Prototype systems (TRL 5–7)** combining sensing hardware, data acquisition, and intelligent analytics, ready for pilot deployment in hydrogen infrastructure. • **Distributed and/or networked detection architectures** enabling spatial localization of hydrogen leaks in complex environments (e.g. refueling stations, pipelines, storage facilities). • New insights into **hydrogen sensing mechanisms**, performance degradation, and cross-sensitivity under realistic environmental conditions. • **Benchmark datasets and performance metrics** for hydrogen leak detection, supporting comparative evaluation of sensing technologies. • **AI-based algorithms** for sensor fusion, false-alarm reduction, and early-warning detection. • Embedded software and edge-computing solutions suitable for industrial deployment. • Operational guidelines for installation, operation, and maintenance of hydrogen leak detection systems. Project has impact at different levels: • Scientific and technological impact: The collaboration advances the **state of the art in hydrogen safety and sensing**, particularly in early-stage and distributed leak detection. TNO's research will de-risk and validate beyond SotA sensors and transfer the technology for the Korean SMEs to industrialize and scale. It bridges the gap between laboratory research and industrial deployment through system-level validation. • Industry impact: The project enables **South Korean SMEs** to accelerate commercialization and access European markets through validated, regulation-ready solutions. Reduces time-to-market and technical risk for new hydrogen safety products. • Societal and policy impact: It builds public and industrial confidence in hydrogen as a clean energy carrier by improving **hydrogen infrastructure safety**, reducing risks of accidents, explosions, and service disruptions. The Dutch Korean collaboration provides evidence-based input to international safety standards and regulatory frameworks and supports the development of a **globally harmonized hydrogen safety ecosystem**.

6. References

(1) Sr. Consultant/ Naval architect Martijn Hoogeland: https://www.linkedin.com/in/martijn-hoogeland-67976617/

(2) Sr. scientist optical instruments Man Xu: https://www.linkedin.com/in/xuman/

(3) TNO patent pending, P135303EP00, “Coating based hydrogen leakage fiber sensor.”

(4) TNO hydrogen sensor LIGH2T, https://ventures.tno.nl/portfolio/ligh2t-large-scale-integrated-gas-h2/

(5) Twerda et al., A numerical study of the behaviour of Hydrogen in ventilated spaces, Advances in Maritime Technology and Engineering, 2024.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization